

MiniSpeedCam r1.1 — PCB Design Review

Radar speed-camera motherboard (ESP32-S3 + STM32F301 + CDM324 24 GHz Doppler) · 4-layer, 50 × 50 mm

Revision: r1.1 **Repo:** Valar-Systems/MiniSpeedCam @ version/r1.1 **Method:** kicad-happy analyzer suite

Audience: Electrical engineering review

94

COMPONENTS

4-layer

F / GND / PWR / B

100%

ROUTED

10 / 100

EMC RISK SCORE

7 / 32

ERRORS / WARNINGS

1. Scope & Method

This review was produced by cloning the open-source [kicad-happy](#) design-review skill suite and running its Python analyzers directly against the **r1.1** KiCad source files (`version/r1.1/hardware/pcb/kicad/`). Findings carry the analyzer's rule IDs for traceability and were triaged by hand against the raw schematic netlist; expected layout artifacts are separated out in §7.

Analyzer	Status	Result
analyze_schematic.py	RAN	94 parts, 101 nets, 37 unique — 1 error / 3 warnings
analyze_pcb.py --full	RAN	4-layer, 50×50 mm, fully routed — 6 errors / 26 warnings
analyze_gerbbers.py	RAN	11 layers complete, PTH + NPTH present — 1 warning
cross_analysis.py	RAN	6 cross-domain findings
analyze_emc.py (44 rules)	RAN	65 findings — EMC risk score 10/100 (scoring is conservative)
Datasheet verification	PARTIAL	Key IC supply/ground pins confirmed against standard pinouts (§6); op-amp signal-pin order is a designer check
analyze_thermal.py	PARTIAL	Ran, but no power-budget input → junction temps not computed (inconclusive, not "clean")
SPICE simulation	SKIPPED	No ngspice / LTspice / Xyce on host — filter/gain values not simulated

Layer stack: F.Cu / In1.Cu = GND plane / In2.Cu = split power plane / B.Cu, 1.6 mm.

2. Executive Summary

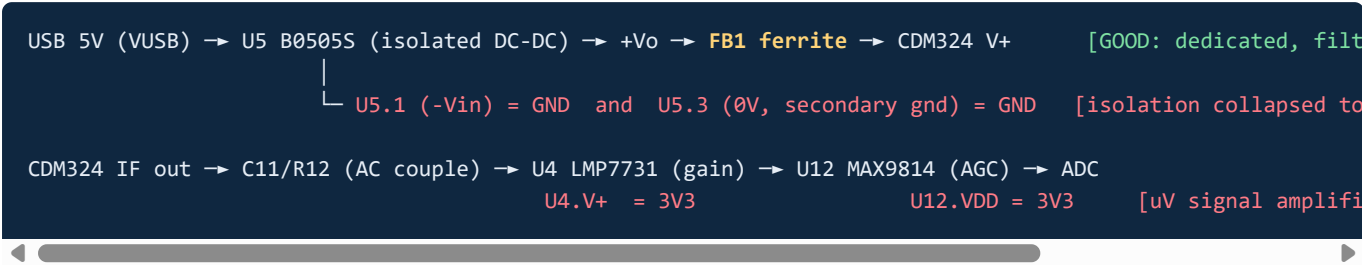
The board is complete, fully routed, and mechanically tidy. The issues that matter cluster around one theme: **the sensitive 24 GHz Doppler analog front-end is not adequately isolated from the digital/RF noise of the ESP32-S3**. This is the electrical root cause behind the firmware workarounds currently in the codebase (a radar-blanking handshake that discards FFT frames during every WiFi burst, plus WiFi TX power deliberately reduced to 11 dBm). A final spin should spend its budget here. The remaining findings are standard pre-fab DFM hygiene (fiducials, edge clearance, test points, BOM MPNs).

Top priority — Give the radar amplifier chain (U4 LMP7731, U12 MAX9814) a dedicated low-noise analog supply and a single-point analog ground, instead of the shared digital 3V3 / GND. This is the change

most likely to remove the need for firmware radar-blanking and let WiFi TX power return to full range.

3. Priority Finding — Radar Front-End Shares the Digital Supply & Ground

Traced from the r1.1 netlist, the radar power/signal architecture is:



Observation	Evidence (netlist)	Severity
Doppler amplifiers on shared digital rail	U4.5 (V+) → 3V3 and U12.5 (VDD) → 3V3 — the same rail as the ESP32-S3 and STM32 logic. A WiFi-TX current transient on 3V3 lands directly in the gain stage. (Both supply pins datasheet-confirmed, §6.)	HIGH
Isolated converter ground bonded to digital ground	U5.1 (-Vin) → GND and U5.3 (0V) → GND ; there is no analog/isolated ground net anywhere in the design (rail list: 1V2, 2V8, 3V3, 5V, GND, VUSB). The B0505S's galvanic isolation is therefore not used — the radar return current shares the digital ground.	HIGH
Diode rail done well	U5.4 (+Vo) → FB1 → J3.3 (CDM324 V+) with C22 bulk — a separate, ferrite-filtered supply for the radar module itself.	OK

Recommended changes

- Derive a clean analog rail for U4/U12 from the isolated B0505S +Vo output via a small **low-noise LDO** (high-PSRR linear regulator), rather than tapping the digital 3V3 .
- Create a local **analog ground island** for the front-end, tied to digital GND at a **single star point** near the STM32 ADC return — keep radar return current off the digital ground pour.
- If the B0505S is kept, decide deliberately whether its secondary 0V joins GND at one point (quiet) or not at all (true isolation). Today it reads as a common-ground bond, which gives up most of the part's benefit.

Note: the VOUT net is the MAX9814 audio output to the ADC, not a supply rail — so the analyzer's RS-001 "VOUT has no declared source" warning is a **false positive**.

4. Ground & Reference-Plane Integrity

These corroborate §3 — a fragmented reference is exactly what couples digital switching into the radar path.

Rule	Severity	Finding & action
PS-002 / GP-001	ERROR	GND resolves into 6 copper islands with 2 signals crossing a gap ; 3V3 into 23 islands , 2 signals crossing. Verify In1.Cu is a continuous ground reference (no signal traces carved through it) and stitch the pours. 3V3 fragmentation matters more than usual here because the analog amps reference 3V3.
RP-001	ERROR	Camera PCLK changes layers with no return-path stitching via ; same for several STM/UART/BOOT0 nets. Add a ground via adjacent to each clock layer-transition.

5. EMC Pre-Compliance & DFM

The EMC score (10/100) is conservative, but the I/O and clock flags are legitimate for a device that radiates near its own radar receiver.

Rule	Severity	Finding & action
IO-001	ERROR	No filtering on J1 (Phoenix power input), USB1, USB2 . Add a TVS + pi/ferrite on the J1 inlet, and a common-mode choke + ESD on each USB-C data pair.
CK-001 / CK-003	WARN	Camera <code>XCLK</code> / <code>PCLK</code> and <code>SD_HOST_CLK</code> on outer layers and routed near connectors J8/J10 — move inner-layer or add guard ground.
DC-003	WARN	13 decoupling caps (C6/C7/C9/C12/C17/C42...) far from their plane via — shorten cap→via path to cut PDN inductance.
FD-001	ERROR	No fiducials on either side (fine-pitch QFN present) — add ≥ 3 global fiducials for assembly.
PM-002	ERROR	C6 and FB1 sit 0.25 mm from the board edge — move in ≥ 0.5 mm (ceramic-cap / ferrite crack risk on depanelization). J1/J3 overhang is intentional edge-mount — ignore.
TE-001	WARN	0 test points . Add TPs on every rail, the radar analog output, and GND scope points — invaluable for bring-up.
DFM-001	WARN	0.1 mm / 0.116 mm trace+space forces JLC advanced tier. Relax non-camera nets to 0.127 mm to drop to standard tier if cost matters.
TV-001	WARN	STM32 (U13) thermal pad has 5/9 recommended vias — top up to 9 (minor).
SS-001	ERROR	MPN coverage 3/37 in the KiCad symbols. Populate MPN fields from the existing <code>MiniSpeedCam_BOM.csv</code> LCSC column — a BOM-hygiene gate, not an electrical defect.

6. Datasheet Verification

The supply/ground facts behind the \$3 priority finding were checked against the parts' documented pinouts:

Part	Pin checked	Result	Basis
U5 B0505S-1W	1 = -Vin, 2 = +Vin, 3 = -Vout(0V), 4 = +Vout	CONFIRMED	Matches standard B0505S 4-pin isolated DC-DC pinout. -Vin and -Vout both on GND verified.
U12 MAX9814	5 = VDD, 6 = MICOUT, 8 = MICIN, GND = 4/7/11/EP	CONFIRMED	Symbol pin map is an exact match to the documented MAX9814 14-pin TDFN pinout. VDD → 3V3 verified.
U4 LMP7731	5 = V+ (positive supply)	CONFIRMED	Pin 5 = V+ for the SOT-23-5 single op-amp. V+ → 3V3 verified.
U4 LMP7731	1-4 = OUT / +IN / -IN order	DESIGNER CHECK	Symbol uses the standard SOT-23-5 single-op-amp order (1=OUT, 2=V-, 3=+IN, 4=-IN). Datasheet figure could not be auto-rendered in this environment — confirm visually against the LMP7731 connection diagram before fab. Not a flagged discrepancy.

No automated datasheet sync was performed; the above are targeted manual confirmations. Other ICs (STM32F301, ESP32-S3-WROOM) use manufacturer reference pin maps and were not exhaustively pin-checked.

7. Triaged as False-Positive / Expected (no action)

- CC-002 (16x) "narrow signal" on camera/I²C nets — 0.1 mm is fine for these low-current signals.
- GR-004 B-paste 26/187 apertures — back side has only 2 components; expected.
- DP-005 USB diff-pair 7% / 1.3 mm length mismatch — negligible at USB full-speed.
- PM-001 courtyard overlaps (U15/C2, U16/C26, ...) — decoupling caps deliberately tucked under their LDOs.
- RS-001 on VOUT — VOUT is a signal (MAX9814 output), not a power rail.

8. Review Gaps & Confidence

- No full datasheet sync.** Supply/ground pins for the front-end parts were spot-verified (\$6); the LMP7731 input/output pin order and the STM32/ESP pin maps were not exhaustively checked against rendered datasheet figures.
- No SPICE simulation** (no simulator on host): front-end gain, AGC behavior, and filter cutoffs were not numerically verified.
- Thermal inconclusive:** no power-budget data, so junction temperatures were not computed. Given the AP63203 buck and small dissipations, thermal is unlikely to be a blocker, but it was not positively cleared.

Bottom line: r1.1 is electrically functional and routable, but its radar receive path is coupled to the digital domain through both supply (3V3) and ground. Fixing that — a dedicated low-noise analog rail + single-point analog ground for U4/U12, a continuous ground reference, and basic I/O filtering — is the highest-value content for a final spin and directly targets the noise problem the firmware is currently working around.